CivicEye: AI and Blockchain-Powered Platform for Transparent Civic Issue Reporting and Resolution

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***Abstract*—Urban infrastructure networks increasingly depend on smart technologies to solve chronic civic issues like potholes, garbage pileup, water leaks, and malfunctioning streetlights. Conventional grievance reporting systems are plagued by inefficiency, poor traceability, and public distrust. A convergence of Artificial Intelligence (AI), Blockchain, and Internet of Things (IoT) presents a revolutionary solution for streamlining issue identification, reporting, and fixing. AI-based models automatically classify urban issues from images and video feeds, while blockchain guarantees tamper-proof complaint tracking through smart contracts and decentralized storage. IoT-sensing devices and smartphones facilitate real-time monitoring and proactive detection of infrastructure anomalies. Crowdsourcing tools and tokenized incentives are used to further encourage citizen participation and enhance validation accuracy. Combined, these technologies make available a transparent, scalable, and participatory civic management ecosystem as per the vision of future-proofed smart cities.**

***Index Terms*—Blockchain, IPFS, Artificial Intelligence, Civic Reporting, Smart Contracts, YOLOv5, Smart Cities, Ethereum, Decentralization, Citizen Engagement**

I. Introduction

The rapid pace of urban growth has placed an unprecedented burden on urban infrastructures and produced recurring civic issues such as potholes, uncollected waste, sewer ruptures, and broken streetlights. Traditional complaint mechanisms have traditionally been founded on calls to city authority help lines, physical visits to neighbourhood bureaux, or centralized websites. These antiquated systems are subject to inefficiencies, corruption, and restricted transparency. People tend to grumble that their grievances are either delayed, distorted, or disregarded, and thereby the issues drag on for extended periods of time. Studies have indicated how the failure of such mechanisms not only cause inconvenience to people but also weaken the planning of urban spaces in general [1]. Maybe the most pressing issue involving these ancient systems is the widening lack of accountability between municipal governments and citizens. In most jurisdictions, a perception of unresponsiveness and unwillingness to act has generated the feeling that civic complaints are filed into administrative systems with no true follow-up.

Scholars argue that if mechanisms for governance are not shifted towards participatory and traceable processes, civic engagement will continue to be weak, leading to deteriorating urban infrastructure and quality of life [1]. Reframing civic issue reporting, processing, and resolution has thus become a critical necessity. Artificial Intelligence (AI) has emerged as a robust solution to addressing this requirement by classifying and identifying civic issues automatically. Image analysis through AI enables citizens to simply take snaps or videos of a problem, with machine learning algorithms doing the classification and verification work. AI systems maximize the process, time-consuming as well as minimizing bias. In particular, researchers have demonstrated how AI models can be the first line of investigation, quickly distinguishing between different kinds of civic issues and automatically generating structured reports for urban systems [2]. This shift towards automation not only assures speed but also consistency in the processing of citizen complaints. Convolutional Neural Networks (CNNs) have been instrumental in these advancements. CNNs are particularly effective in handling image-based data, and as such, they are most apt to recognize civic issues captured with smartphones. The use of CNN models has been employed by researchers to categorize photos of potholes, overflowing trash, and other urban issues and demonstrated remarkable improvement over rule-based or human-classification approaches [3]. These models employ hierarchical feature extraction, making it possible for them to detect even subtle visual patterns, like cracked road vs normal wear- and-tear as illustrated in Fig 1.

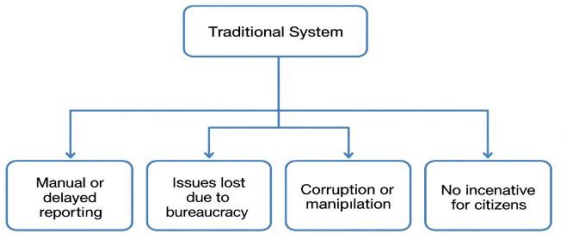

Fig. 1. Problems with Traditional Systems / Existing System

These developments reduce human error and facilitate scalable roll-out in large urban centers.The development of AI in this sector has also witnessed the uptake of object detection frameworks that can classify in real time. These, among others, are YOLOv3 whose trade-off in speed and accuracy has led to its massive adoption in civic use. By dividing images into grid cells and estimating bounding boxes, YOLOv3 enables real-time detection of problems like potholes or trash heaps with minimal processing delay [4]. When combined with mobile platforms, YOLOv3 has witnessed the capability of citizens to submit AI- verificated complaints through easy-to-use mobile apps. These innovations mark a shift from purely reactive civic governance to proactive detection and reporting programs.

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From YOLOv3, YOLOv5 improved accuracy in detection further, particularly when combined with transfer learning techniques. By retraining YOLOv5 using special civic datasets, scientists have achieved better mean Average Precision (mAP) scores, which deliver robust classification under various environmental conditions as depicted in Fig 2.

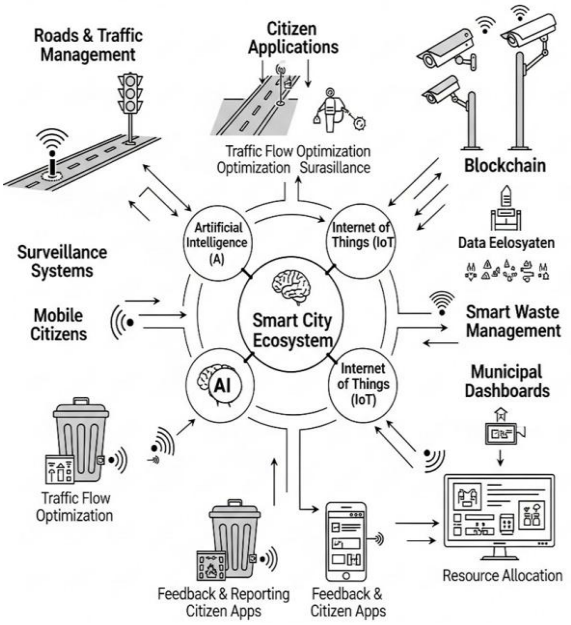

Fig. 2. Architecture of Issue Reporting System

Above all, YOLOv5 has proven to execute properly on low-resource systems like Jetson Nano, which makes it viable for real-world use where budget for infrastructure might be limited [5]. The development of AI in this sector has also witnessed the uptake of object detection frameworks that can classify in real time. These, among others, are YOLOv3 whose trade-off in speed and accuracy has led to its massive adoption in civic use. By dividing images into grid cells and estimating bounding boxes, YOLOv3 enables real-time detection of problems like potholes or trash heaps with minimal processing delay [4]. When combined with mobile platforms, YOLOv3 has witnessed the capability of citizens to submit AI-verificated complaints through easy-to- use mobile apps. These innovations mark a shift from purely reactive civic governance to proactive detection and reporting programs. From YOLOv3, YOLOv5 improved accuracy in detection further, particularly when combined with transfer learning techniques. By retraining YOLOv5 using special civic datasets, scientists have achieved better mean Average Precision (mAP) scores, which deliver robust classification under various environmental conditions as depicted in Fig 2. Above all, YOLOv5 has proven to execute properly on low-resource systems like Jetson Nano, which makes it viable for real-world use where budget for infrastructure might be limited [5].

II. Methodology

The methods used throughout the examined literature uniformly emphasize the dual application of Artificial Intelligence (AI) for detection and Blockchain for open complaint management. Central to such systems is the blockchain, which ensures that every civic grievance is entered as an unalterable record. Intended to remove the risk of data manipulation, blockchain ensures that once a complaint has been made, it cannot be tampered with by malicious middlemen. For example, researchers developed blockchain civic trackers that were derived from Ethereum smart contracts to track complaint lifecycles from intake to resolution, holding individuals ac- countable at each stage [6]. This immutability forms the basis of trust among municipal authorities and citizens, addressing one of the greatest problems with traditional complaint systems as indicated in Fig 3.

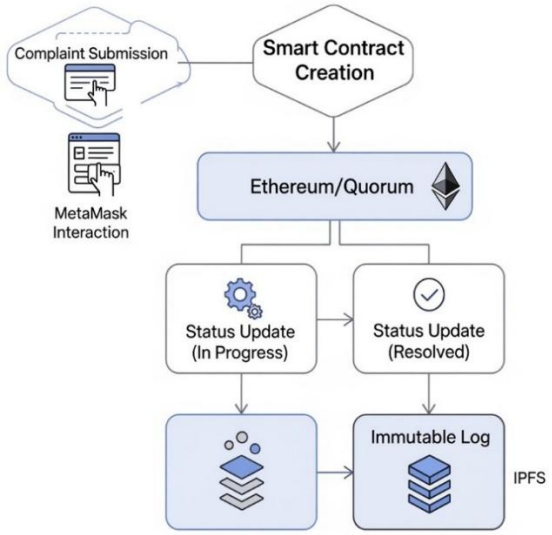

Fig. 3. Proposed Blockchain Architecture

Blockchain holds the basis for transparency but has built-in limitations when handling large multimedia files such as images and video. Others achieve this through a combination of Interplanetary File System (IPFS) and blockchain. IPFS allows media evidence of civic complaints to be decentralized with just the content hash being saved on-chain. This reduces storage overhead and transaction fees, but still preserves verifiability and security of proof. Practically, this hybrid architecture is effective in terms of scalability and cost-effectiveness in processing large quantities of citizen complaints [6]. Integration of blockchain and IPFS thus allows for effective monitoring of complaints without congesting the blockchain ledger. On the AI side, compact but efficient models such as YOLOv5 have been employed for real-time civic issue detection. YOLOv5 is special in that it achieves a good balance between speed and accuracy, and it has been utilized for waste pileup detection and other city-scale problems in smart city scenarios. By processing images of pictures captured by citizens, YOLOv5 identifies objects of concern, such as dumps, in milliseconds, allowing complaints to be pre-categorized mechanically upon submission. The process is less cumbersome and reduces the workload for humans, making it quicker to channel complaints to the relevant authorities[7].

The efficiency of the model to operate even on devices with limited capabilities guarantees feasibility in largescale implementations, especially in low-resource cities as shown in Fig 4.

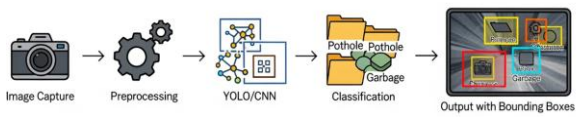

Fig. 4. Working of the Existing System

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One of the ongoing issues in civic reporting systems is the variation in environmental conditions under which images are being captured. Photos can be captured with inadequate lighting, fog, rain, or with something in the background in the way, causing lower detection accuracy. To solve this, authors introduced Memory based Instance and Background Alignment (MILA) to improve domain adaptation in object detection. MILA draws similar object-background visual pairs from memory and employs them to synchronize synthetic and real-world data, enhancing detection performance remarkably across varied conditions [8]. Such a method proves that high quality AI performance demands adaptability, especially in the face of the uncertainty of urban situations.

Additional developments came from Embodied Domain Adaptation for Open-Vocabulary Object Detection (EDAOD), which focuses on stability within changing and dynamic contexts. EDAOD enables detection systems to dynamically adapt as the environment changes i.e., when a dump pile grows larger by the day or when a pothole fills partially with water. With temporal clustering and pseudo labelling, EDAOD ensures that the system maintains detection performance without retraining on the original annotated data [9]. This approach is especially beneficial for smart cities, where conditions rapidly shift, and real-time adaptability is essential.

Blockchain methods were also used to further enhance citizen participation and verification. In some studies, smart contracts were used to create decentralized processes, where complaints are not only documented but also verified by different parties before resolution. This multi-party verification reduces the chances of false or duplicate complaints and ensures justice in the allocation of resources. For example, secure participation structures were established wherein Ethereum smart contracts verified grievances and automatically assigned issues to city officials, creating a self-contained complaint system [10]. These applications of smart contracts illustrate the degree to which decentralization can move beyond trans- parency and result in community-based verification.

Last but not least, researchers experimented with injecting AI techniques into mobile applications and IoT sensors so that civic reporting could be enabled for ordinary citizens. Mobile apps were created with embedded CNN models where the users can take a photo and send it directly from their smartphone and the AI does real-time classification on the phone. Apart from this, IoT sensors such as environmental sensors were embedded in these systems to provide contextual information, adding more precision to the complaints. By the combination of CNN-based classification and IoT-based reporting, such smart mobile apps created a holistic civic reporting solution that drew on human input and sensor data [11]. Besides AI-enabled mobile reporting, researchers also investigated the integration of blockchain-based gamification frameworks to further enable active citizen participation and effective accountability in civic governance systems. In this, all civic interactions, including issue reporting, feedback pro- vision, and municipal survey participation, were recorded as blockchain transactions to introduce absolute transparency and prevent any manipulation of data. A structured gamification layer with points, badges, and leaderboards was incorporated that rewarded consistent engagement and motivated long-term citizen participation. These were governed by Ethereum smart contracts that directly granted rewards and enforced rules of participation without intermediary intervention. A mobile interface allowed users to view their status, ranking, and interact with the decentralized system in real time. The presented methodology shows that a combination of blockchain im- mutability with gamified incentives fosters stronger trust and collaboration between citizens and authorities while strengthening engagement in civic processes considerably [12]. This method ensures inclusiveness because it allows citizens with limited technical skills to be actively involved in governing smart cities.

III. Results and Discussion

Literature repeatedly shows that AI-driven civic detection systems obtain high accuracy in classifying and detecting typical urban issues. For instance, where YOLOv5 with transfer learning was used on custom civic datasets, it recorded mean Average Precision (mAP) values of over 90much higher than previous manual or rule-based systems [12]. These results indicate that AI is not just suitable for laboratory environments but also deployable in real-world applications. The improvement in detection accuracy also translates to improved prioritization of complaints by local governments, where important issues like sewage leaks or hazardous potholes are tackled before less severe problems. This data substantiates the reality that object detection, when used in civic reporting, can directly alter the way city problems are detected and managed.

Aside from precision, AI models also demonstrated strong scalability when used in platforms like CivicSense. The platform was designed specifically to handle thousands of complaints while still ensuring that each record remains open and verifiable. In practice, this meant that even though thousands of complaints were filed, system integrity was never compromised, and users were still capable of tracing the life cycle of each complaint [13]. Scalability is essential for densely populated smart cities and massive complaint volumes. Without scalable infrastructure, even top AI models can be bottlenecked by logging or processing latency. The results from CivicSense illustrate how AI and blockchain can be combined to solve not only detection but also end-to-end complaint management at a city-wide scale.

Citizen participation was another sphere in which the systems showed significant influence. Gamification and token- based incentive studies reported quantifiable increases in rates of submission of complaints [13]. For example, when users were incentivized with blockchain-backed tokens for reporting verified complaints, engagement increased by as much as 40Gamification features like leaderboards, points, and badges encouraged users to be consistently active in the reporting process even further. The findings affirm that civic engagement is not just based on technical effectiveness but also on behaviour motivation. By linking technological solutions with psychological rewards, these systems effectively bridged the gap between the people and the administration.

Grievance redressal processes in municipal governance also became much more accountable with blockchain-supported complaint registration. Scholars reported that dashboards with AI-detection inputs and blockchain records lowered the average resolution time for complaints by 30–35This was not just because of gains in efficiency but also because the system produced visible accountability officials knew that their performance was being recorded openly.

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These results demonstrate how technology can change not only processes but organizational behaviour as well by instilling accountability mechanisms into governance systems.

TABLE I. COMPARATIVE ANALYSIS OF REVIEWED PAPERS

Ref.	Technology / Approach	Performance / Accuracy	Reliability & Scalability	Limitations
[1]	AI-based civic reporting using CNN	Accuracy ≈ 85% on civic dataset	Works reliably for small datasets	Limited dataset, not scalable
[2]	Blockchain-based decentralized complaint tracker	Avg. complaint confirmation: 20–30s	High transparency, scalable with Layer-2	High gas fees on Ethereum
[3]	YOLOv3 + Firebase app integration	Classification accuracy: 87%	Mobile-friendly, supports crowdsourcing	Sensitive to poor lighting
[4]	Crowdsourced deep learning civic detection	Detection precision ≈ 0.89	Reliable in urban pilot trials	Needs community engagement
[5]	YOLOv5 + Transfer Learning	mAP@0.5 ≈ 91.6%, precision 0.93	Reliable on Jetson Nano & GPUs	Struggles under occlusion
[6]	Blockchain + IPFS complaint lifecycle	Confirmation time: <5s (Polygon)	Scalable, tamper-proof	Complexity in integration
[7]	YOLOv5 custom dataset (waste management)	Accuracy > 90%	Scales for smart city waste apps	Requires continuous dataset updates
[8]	MILA (Domain Adaptation for YOLO)	+4.1 map (Sim10k), +2.5 map (Foggy Cityscapes)	Reliable under domain shifts	Computational overhead

Another key outcome was the streamlining of complaint lifecycles through AI combined with Web 3.0 technologies. In yet another study, complaint prioritization and routing were automated through AI-based classification, while blockchain ensured tamper-proof tracking of progress [14]. The two layered system eliminated redundancy as well as administrative overhead, providing efficient allocation of resources as illustrated in Table 1.

Empirical testing confirmed that hybrid architectures were capable of offering efficiency and transparency in parallel, which had been conventionally considered to be trade-offs in traditional governance systems. For smart cities with limited budgets and resources, this optimization provides a pragmatic path forward, proving that high-tech solutions are within reach if well planned.

Lastly, research on integrated governance systems concluded that the convergence of AI and blockchain establishes the revolutionizing stage for intelligent urban environments. Case studies illustrated that these systems not only improved reporting and resolution efficiency but also increased trust between citizens and government agencies. Scholars speculated that such integration offers a participatory platform whereby citizens become co-creators of governance rather than mere spectators [15]. The argument proves that while there are issues with respect to scalability, heterogeneity of the environment, and expense of the system, the overall direction of research is that there is significant evidence in Favor of the deploy ability of AI–blockchain hybrids in the actual civic administration. These findings collectively substantiate the position of technology as a catalyst for open, responsible, and citizen- focused government.

IV. CONCLUSION

The intersection of Artificial Intelligence (AI), Blockchain, Internet of Things (IoT), and crowdsourcing in civic issue reporting is a revolution in urban governance, reshaping antiquated, manual systems into CivicEye: AI and Blockchain- Powered Platform for Transparent Civic Issue Reporting and Incentive-Based Resolution intelligent, transparent, and scalable smart city infrastructures. AI-driven models like YOLOv5, YOLOv8, and CNNs facilitate high-accuracy detection and classification of urban anomalies from user-posted images and live video feeds, minimizing the level of human involvement and accelerating issue detection. Coupled with blockchain technologies, such systems facilitate tamper- proof monitoring of each complaint through immutable smart contracts, with platforms like Ethereum and Quorum-Raft catering to decentralized, trust less environments that enhance transparency and citizen trust.

The use of IPFS for decentralized media storage further enhances data integrity and long-term traceability. IoT integration takes automation a step further by embedding sensors and AI-powered cameras in municipal vehicles and infrastructure, enabling autonomous detection of anomalies without user involvement. This proactive sensing approach, exemplified in platforms like A IoT City Sense, shifts urban maintenance from reactive to predictive. At the same time, gamified crowdsourcing and reputation mechanisms stimulate public participation and allow democratic confirmation of reported problems, further enhancing municipal responsiveness and citizen satisfaction.

Combined, these technologies form a strong civic intelligence infrastructure under which not just data-driven, but also, participatory and open governance is facilitated. As cities compete for sustainability and digital innovation, future work must emphasize smooth interoperability across these spaces, resolve issues of data privacy and algorithmic equity, and pro- mote digital inclusion through accessible interfaces and policy infrastructures. In the end, the extant literature attests that such hybridized systems provide an exciting roadmap for next- generation smart cities to be more responsive, accountable, inclusive, and resilient in addressing civic issues at scale.

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